Confidence Is Not Coverage: Magnitude-Dependent Miscoverage and Conformal Recalibration of Prediction Intervals in Deployed Equity Forecasting

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Abstract

Probabilistic forecasters are increasingly deployed to quote not only a point prediction but a calibrated interval that a downstream actor treats as a risk budget. Whether those intervals keep their promise in production is rarely studied. We present an empirical study of the interval reliability of a deployed equity-forecasting service whose every forecast — a median price together with a nominal 90% interval — is written to an append-only, point-in-time ledger before the outcome is known. On 619 resolved forecasts the intervals attain only 70.6% empirical coverage, confirming that deployed uncertainty is overconfident and not self-correcting. The novel finding is that the miscoverage is *not* uniform and, importantly, is not explained by interval width: coverage is essentially flat across width quartiles (0.68–0.73) yet collapses with the magnitude of the point forecast, from 95.7% on the least opinionated third of forecasts to 47.1% on the most opinionated third. The system is least reliable exactly when it is most confident — a *confidence–coverage inversion* — because the interval generator fails to widen for aggressive forecasts (the correlation between interval width and forecast magnitude is only 0.27). We then show that a minimal split-conformal multiplicative width recalibration, fitted on a past calibration window and applied walk-forward, restores marginal coverage to 94.4% on held-out forecasts and lowers the mean interval (Winkler) score from 0.218 to 0.185. A magnitude-conditional variant recovers the correct, oppositely-signed corrections (shrink the easy intervals, widen the confident ones) but does not beat the global fix on the present two-month, six-name sample; we argue that conditioning is the right structure but is not yet warranted by the data, and we state precisely the evidence that would warrant it. The contribution is a reproducible, verifiable methodology for diagnosing and correcting interval reliability in production, together with a cautionary empirical law: in a near-efficient forecasting regime, confidence and coverage move in opposite directions unless the interval explicitly scales with forecast magnitude.

Keywords: prediction intervals; conformal prediction; calibration; uncertainty quantification; interval reliability; financial forecasting; machine learning deployment.

1. Introduction

A growing number of machine-learning systems are deployed not to classify but to *quantify uncertainty*: they emit a point prediction together with an interval that a human or an automated policy treats as a budget for risk. A weather model states a temperature and a band; a demand forecaster states a quantity and a band; an equity model states a price and a band. In each case the interval, not the point, is what makes the output actionable, because the consumer sizes an action against the stated uncertainty. A 90% interval is a promise: across many such forecasts, the realized value should fall inside the stated band nine times in ten. When the promise is kept, the interval is a contract the consumer can plan against. When it is broken — when the bands are systematically too narrow — every downstream risk calculation is silently wrong, and the error is invisible precisely because the point prediction can look reasonable.

Whether deployed intervals keep their promise is rarely measured in the open. Most reported evaluations of probabilistic forecasters are computed on a backtest: the same historical sample, often the same cross-validation folds, used to build the model. Backtests are necessary but they are not the deployment. A model that is well calibrated in cross-validation can drift out of calibration in production because the live distribution shifts, because the calibration set was not representative, or simply because the act of deploying changes what data arrive [1,2]. The honest test of an interval is an analysis on forecasts that were recorded *before* their outcomes were known, never revised, and resolved against the realized value. That is the test we run here.

We study a deployed equity-forecasting service for liquid Indian (NSE) equities. The service issues, at each decision time, a median price forecast at a fixed horizon together with a nominal 90% price interval, and it appends the forecast to a write-once ledger keyed by issue time. When the horizon elapses the realized price is joined to the record. The ledger is therefore a point-in-time, leakage-free substrate for exactly the analysis described above. On the 619 forecasts resolved during a two-month window we ask one question: *do the* 90% *intervals cover* 90% *of the time, and if not, what explains the gap?*

The headline answer — that the intervals attain only 70.6% coverage — is, by itself, an unsurprising instance of deep-model overconfidence [1,3]. The contribution of this paper is what lies underneath that number. We decompose miscoverage along the covariates a practitioner would reach for and find a sharp, counter-intuitive structure:

- Coverage is essentially *flat* across interval-width quartiles $(0.72, 0.68, 0.69, 0.73)$: wider intervals are not better covered, so width alone does not diagnose the failure.
- Coverage *collapses with the magnitude of the point forecast*, falling monotonically from 95.7% on the least opinionated third of forecasts to 47.1% on the most opinionated third.

The system is, in other words, least trustworthy exactly when it is most confident. We call this the *confidence–coverage inversion*. Its mechanism is concrete: the interval generator barely widens when the model forecasts a large move (the correlation between normalized interval width and forecast magnitude is only 0.27), so an aggressive forecast is paired with a band scarcely wider than a timid one, and the band is blown through whenever the market reverts — which, near the efficient-market limit, is most of the time.

Having diagnosed the failure we correction it with the lightest possible instrument. We apply a split-conformal multiplicative recalibration that scales every interval's half-width by a single factor estimated on a past window, and we evaluate it walk-forward on later, unseen forecasts. The factor $\lambda^\star \approx 1.54$ restores marginal coverage to 94.4% and improves the mean interval score from 0.218 to 0.185 on held-out data. We then test the obvious refinement suggested by the diagnosis — conditioning the factor on forecast magnitude — and report a deliberately negative result: the conditional fit recovers the correct, oppositely-signed corrections (a factor below one for timid forecasts, near two for confident ones) but does not improve the coverage–sharpness trade-off over the global fix on the present sample. We take the restraint seriously: conditioning is the right structure, but the two-month, six-name ledger does not yet contain enough confident-forecast outcomes to estimate it reliably, and we say exactly what would.

Our contributions are:

1. **An evaluation procedure.** A reproducible, point-in-time procedure for testing whether a deployed forecaster's intervals keep their nominal promise, using a write-once ledger so that no outcome can leak into the evaluation (Section 3).

2. **The confidence–coverage inversion.** Empirical evidence that, in a near-efficient regime, interval miscoverage is governed by forecast magnitude rather than by interval width or horizon, with coverage falling from 95.7% to 47.1% across magnitude tertiles (Section 4).
3. **A conformal correction and an honest ablation.** A minimal multiplicative width recalibration that restores coverage walk-forward, plus a magnitude-conditional variant whose benefit we decline to claim on the present sample, with a stated criterion for when it should be adopted (Sections 5–6).

Throughout we are explicit about what the data can and cannot support. The point forecasts in this system carry no directional edge beyond drift — the realized sign agrees with the forecast sign only 45.7% of the time — so the interval, not the point, is the product, and interval reliability is the property worth getting right. Section 7 returns to why this restraint is the correct default for outward-facing systems.

2. Related Work

Calibration and overconfidence. Modern neural and ensemble predictors are routinely miscalibrated, typically overconfident, and the gap is not removed by better accuracy [1–3]. Post-hoc remedies for classification probabilities include Platt scaling [4], isotonic regression [5], and temperature scaling [1], with reliability summarized by the expected calibration error and the Brier score [6, 7]. This literature largely concerns class-probability calibration; our object is different — the coverage of a real-valued *prediction interval* — and the failure mode we isolate (miscoverage indexed by forecast magnitude rather than by a probability bin) has no direct analogue in the temperature-scaling picture.

Conformal prediction. Conformal methods construct intervals with distribution-free marginal coverage guarantees under exchangeability [8–10]. Conformalized quantile regression (CQR) calibrates model-emitted quantiles to achieve coverage [11], and adaptive variants extend the guarantee to distribution shift and time series [12, 13]. Our recalibration is deliberately the simplest member of this family — a scalar multiplicative adjustment of an existing interval, equivalent to a one-parameter conformal calibration of a symmetric nonconformity score — chosen because the contribution is the *deployment analysis and the conditional structure of miscoverage*, not a new estimator. The marginal guarantee of split conformal is exactly what lets us state honestly when the magnitude-conditional refinement is and is not justified.

Proper scoring and sharpness. Forecast intervals should be assessed by proper scoring rules that reward both calibration and sharpness, not by coverage alone [14, 15]. We use the interval (Winkler) score [16], which penalizes width plus a coverage-failure term, so that "cover everything by making the band enormous" is correctly scored as a poor forecast. The principle of maximizing sharpness subject to calibration [15] is the lens through which we read the coverage–width trade-off.

Evaluation integrity in financial machine learning. A long methodological line warns that financial ML results are fragile under backtest overfitting, multiple testing, and look-ahead leakage [17–19], and that near-efficient markets leave little exploitable single-name directional signal [20–22]. Deep and ensemble forecasters for markets are now standard [23, 24], but reports rarely examine whether the quoted *uncertainty* is trustworthy in deployment. Our work is complementary: we take the point forecast's lack of edge as given and ask whether the surrounding interval is honest. The append-only, point-in-time ledger we use to do so is in the spirit of recent calls for living, leakage-free evaluation of deployed models [2, 19].

Table 1: Study sample: resolved forecasts with well-formed 90% intervals.

Property	Value
Resolved forecasts (n)	619
Distinct instruments	6
Issue window	2026-04-19 to 2026-06-12
Nominal interval level	0.90 (all)
Horizon = 10 d / 20 d / 5 d	573 / 38 / 8
Overall empirical coverage	0.706
Median normalized width w	0.114
Forecast-sign accuracy (center vs. realized)	0.457

3. The Deployed System and Its Evaluation Ledger

3.1. Forecasts

The system under study serves liquid NSE equities. At a decision time t for an instrument with price P_t (the *anchor*), it emits a median forecast price $\hat{P}_{t+h}$ at horizon h trading days, and a nominal $1-\alpha = 0.90$ price interval $[L_t, H_t]$. We define the *forecast return* $f_t = (\hat{P}_{t+h} - P_t)/P_t$, the realized return $r_t = (P_{t+h} - P_t)/P_t$, the interval *center* $c_t = (L_t + H_t)/2$ and *half-width* $hw_t = (H_t - L_t)/2$, and the *normalized width* $w_t = (H_t - L_t)/P_t$. The point forecast is produced by a hybrid ensemble; the precise architecture is not needed for what follows, because the analysis treats the forecaster as a black box that emits $(\hat{P}, L, H)$ and asks only whether the band is honest.

3.2. The append-only ledger

Every forecast is written, at issue time, to a write-once row containing the instrument, issue timestamp, target date, horizon, anchor price, median forecast, interval bounds, nominal level, and model version. The row is immutable. When the horizon elapses, a separate resolution pass joins the realized price and the binary hit indicator. Because the predictive fields are sealed before the outcome exists, the ledger is structurally incapable of look-ahead leakage: no quantity used to score a forecast can have been influenced by its outcome. This is the property that makes the analysis credible, and it is a design choice we recommend independent of our specific findings.

3.3. Study sample

We analyze every resolved forecast carrying a well-formed interval issued between 19 April and 12 June 2026: $n = 619$ forecasts across six instruments. All carry the same nominal level (0.90). Horizons are dominated by the ten-day swing setting (573 forecasts), with smaller twenty-day (38) and five-day (8) populations. Table 1 summarizes the sample. Two caveats are stated up front and revisited in Section 8: the window is short (about eight weeks) and the cross-section is narrow (six names), so the analysis characterizes *this* deployment in *this* regime, not equities in general. We make no claim beyond it.

3.4. Novelty of the present study

To our knowledge, the *conditional* coverage of a deployed equity forecaster's price intervals — in particular its dependence on the magnitude of the point forecast — has not previously been characterized on a live, point-in-time ledger. The confidence–coverage inversion reported below is, as far as we are aware, new, and the multiplicative width recalibration we use is orthogonal to the separate question of how often a calibrator should be refreshed: it determines *by how much* the intervals must be rescaled, not *when*.

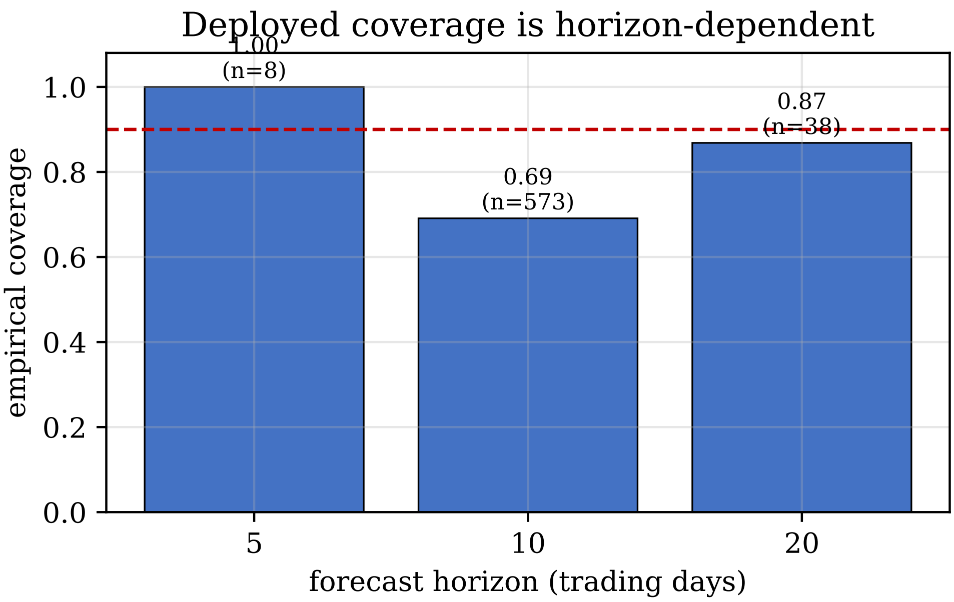

Figure 1: Empirical coverage by forecast horizon (dashed line: nominal 0.90). The ten-day stratum, which holds 573 of 619 forecasts, carries the shortfall; the sparse longer horizons are mildly conservative.

4. Coverage Diagnosis

We write the coverage indicator $\mathbb{1}\{L_t \leq r'_t \leq H_t\}$ where $r'_t = P_{t+h}$ is the realized price, and report the empirical coverage $\widehat{C} = \frac{1}{n}\sum_t \mathbb{1}\{L_t \leq P_{t+h} \leq H_t\}$ overall and within strata.

4.1. The intervals are overconfident

Marginal coverage is $\widehat{C} = 0.706$ against a nominal 0.90 — a 19.4 percentage-point shortfall. Roughly three forecasts in ten fall outside a band that promised to contain nine in ten. Median normalized width is 0.114, i.e. a typical ten-day band spans about $\pm 5.7\%$ around the anchor. The shortfall is not a small finite-sample wobble: a one-sided binomial test of $C \geq 0.90$ is rejected at any conventional level. Overconfidence of this magnitude is consistent with the broader finding that deployed predictors understate their uncertainty [1,2]; the interesting question is its internal structure.

4.2. Horizon explains little

Coverage by horizon is 0.69 at ten days ($n = 573$), 0.87 at twenty days ($n = 38$), and 1.00 at five days ($n = 8$); median widths scale with horizon as expected $(0.111, 0.168, 0.088)$. Figure 1 shows the pattern. The dominant ten-day population drives the marginal shortfall, while the sparse longer-horizon bands are, if anything, slightly conservative. Because the ten-day stratum is where essentially all the data and all the miscoverage live, horizon is not the lever that explains the failure; we therefore set it aside and look within the ten-day-dominated sample.

4.3. Width does not explain miscoverage

A natural hypothesis is that the narrow intervals are the uncovered ones — that coverage simply tracks width. It does not. Splitting forecasts into normalized-width quartiles yields coverage $0.72, 0.68, 0.69, 0.73$ from sharpest to widest (Figure 2a). Coverage is flat in width: the widest quartile is no better covered than the sharpest. Whatever is breaking the intervals is not captured

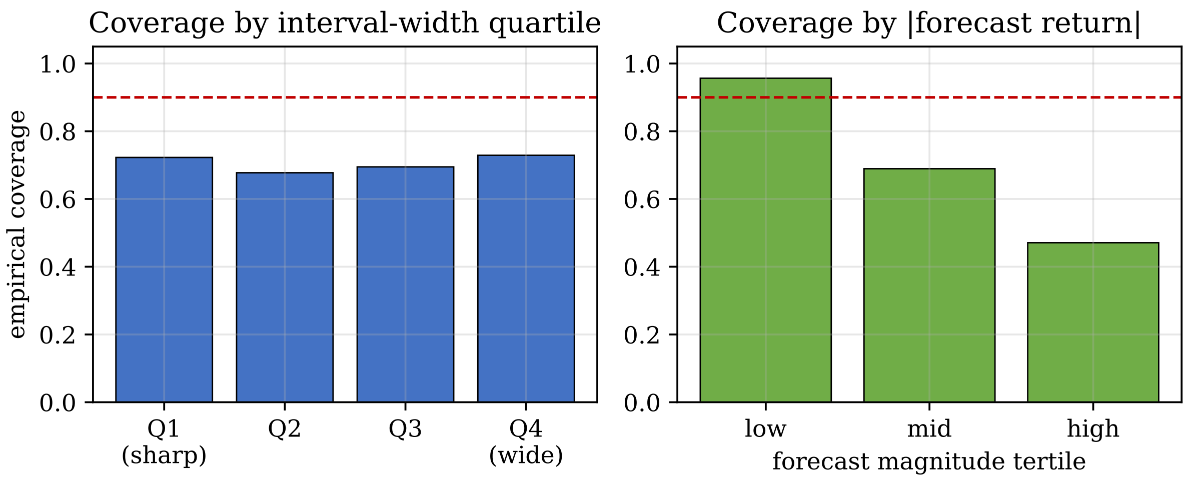

Figure 2: Conditional coverage. **(a)** Coverage is flat across interval-width quartiles. **(b)** Coverage collapses with forecast magnitude, from 95.7% on the least opinionated third of forecasts to 47.1% on the most opinionated third. Dashed line: nominal 0.90.

by how wide they happen to be.

4.4. The confidence–coverage inversion

The covariate that *does* explain miscoverage is the magnitude of the point forecast. Partitioning forecasts into tertiles of $|f_t|$ gives coverage 0.957 (low, mean $|f| = 0.6\%$), 0.689 (mid, 1.9%), and 0.471 (high, 4.3%): coverage falls monotonically and steeply as the model becomes more opinionated (Figure 2b). On the third of forecasts where the model predicts the largest moves, fewer than half of the realized prices land inside the 90% band. The system is least reliable exactly where it is most confident.

The mechanism is that the interval generator does not widen enough for confident forecasts. The correlation between normalized width w_t and forecast magnitude $|f_t|$ is only 0.27; the median width grows from 0.107 on the low-magnitude tertile to just 0.129 on the high-magnitude tertile — a 20% increase in width to accompany a roughly seven-fold increase in forecast magnitude. An aggressive forecast is thus quoted with a band scarcely wider than a timid one. Because the point forecasts carry no directional edge (the realized sign matches the forecast sign only 45.7% of the time, below a coin), large forecasts are followed by reversion as often as not, and a band that failed to widen is blown through. The inversion is the interval-level shadow of an over-extrapolating point forecast.

A small but telling corollary: 94.0% of the intervals *straddle the anchor* price, i.e. they do not commit to a direction at the 90% level. The system already "knows," through the width of its bands, that it usually should not commit. The failure is concentrated in the minority of cases where it does commit — and there it under-covers most.

4.5. Heterogeneity across instruments

Per-instrument coverage (names with ≥ 10 resolved forecasts) ranges from 0.46 to 0.94, mean 0.68. The dispersion is wide enough that a single marginal number hides instruments whose intervals are badly broken and others that are nearly honest. We return to instrument conditioning, alongside magnitude conditioning, when we ask what the data can support in Section 6.

5. A Minimal Conformal Recalibration

5.1. Multiplicative width recalibration

We correction the bands with the smallest instrument that the diagnosis licenses: a scalar multiplicative rescaling of each interval's half-width about its own center. For a factor $\lambda > 0$ the recalibrated interval is

$$[\, c_t - \lambda\, hw_t,\; c_t + \lambda\, hw_t \,]. \tag{1}$$

Define the symmetric nonconformity score $\rho_t = |P_{t+h} - c_t|/hw_t$, the number of half-widths by which the realized price misses the center. The recalibrated interval covers iff $\rho_t \leq \lambda$, so empirical coverage at multiplier λ is exactly the empirical CDF of ρ evaluated at λ:

$$\widehat{C}(\lambda) = \tfrac{1}{n} \textstyle\sum_t \mathbb{1}\{\rho_t \leq \lambda\}. \tag{2}$$

This makes the entire coverage–width trade-off a single monotone curve (Figure 3), and the choice of λ a one-dimensional conformal calibration.

Definition 1 (Split-conformal multiplier)**.** *Given a calibration set $\mathcal{C}$ of size m, the conformal multiplier at level $1-\alpha$ is the $\lceil (m+1)(1-\alpha) \rceil$-th smallest value of $\{\rho_t : t \in \mathcal{C}\}$. Applied to an exchangeable test forecast, it yields marginal coverage at least $1-\alpha$ [8, 11].*

5.2. Global versus magnitude-conditional

The diagnosis suggests conditioning λ on forecast magnitude. We therefore compare two estimators, both fitted only on the calibration window:

- **Global:** one multiplier $\lambda^\star$ from all calibration scores.
- **Magnitude-conditional:** a separate multiplier $\lambda^\star_b$ for each tertile b of $|f_t|$, applied to test forecasts according to their own magnitude.

We deliberately do *not* fit per-horizon or per-instrument multipliers on this sample: with most strata holding only tens of forecasts, a stratified multiplier is an extreme order statistic of a tiny set and would overfit. The same caution governs how we read the magnitude-conditional result.

5.3. Protocol and metrics

Forecasts are ordered by issue time and split 60/40 into a past calibration window ($m = 371$) and a future test window (248); multipliers are fit on the former and all coverage and sharpness numbers are reported on the latter, so the evaluation is strictly walk-forward. We report empirical coverage, median normalized width (sharpness), and the normalized interval (Winkler) score

$$S_\alpha = \frac{(H-L) + \frac{2}{\alpha}(L-y)\mathbb{1}\{y<L\} + \frac{2}{\alpha}(y-H)\mathbb{1}\{y>H\}}{P}, \tag{3}$$

which is a proper scoring rule (lower is better) rewarding sharpness subject to coverage [14,16]. We summarize conditional calibration by the mean absolute coverage gap across magnitude tertiles, $\overline{\Delta} = \frac{1}{3}\sum_b |\widehat{C}_b - 0.90|$.

6. Results

6.1. Global recalibration restores coverage

On the held-out test window the raw intervals cover 0.718. The global conformal multiplier estimated on the calibration window is $\lambda^\star = 1.54$: the bands must be widened by about half

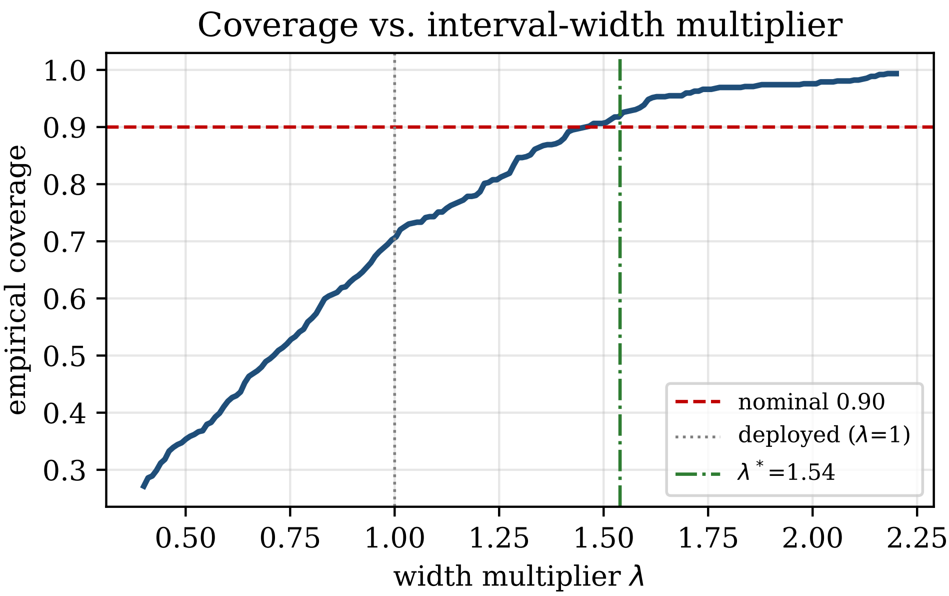

Figure 3: Coverage as a function of the width multiplier λ (whole sample). The deployed bands ($\lambda = 1$) cover 70.6%; the conformal multiplier $\lambda^\star = 1.54$ reaches the nominal 0.90. The curve is the empirical CDF of the nonconformity score ρ.

Table 2: Walk-forward recalibration on the held-out test window (248 forecasts). Coverage target 0.90; interval score lower is better; $\overline{\Delta}$ is the mean absolute coverage gap across magnitude tertiles.

Method	Coverage	Median w	Int. score	$\overline{\Delta}$
Deployed (λ=1)	0.718	0.109	0.218	0.156
Global ($\lambda^\star$=1.54)	0.944	0.168	0.185	0.071
Magnitude-conditional	0.980	0.195	—	0.073

to keep their promise. Applied walk-forward it lifts test coverage to 0.944 and lowers the mean interval score from 0.218 to 0.185 (Table 2). The improvement in a *proper* score is the important part: coverage is restored not by indiscriminate widening (which a proper score would penalize through the width term) but by a correction that the score certifies as a better forecast. Figure 3 shows the full coverage–λ curve; the deployed system sits at $\lambda = 1$ with 0.706 coverage, and the nominal target is reached at $\lambda^\star$.

6.2. An honest negative result on conditioning

The magnitude-conditional fit recovers exactly the corrections the diagnosis predicts: its tertile multipliers are $\lambda^\star_{\text{low}} = 0.89$, $\lambda^\star_{\text{mid}} = 1.40$, $\lambda^\star_{\text{high}} = 2.02$. It *shrinks* the timid intervals (which were over-covering) and roughly *doubles* the confident ones (which were badly under-covering). This is the structurally correct behavior, and it is reassuring that an unsupervised conformal fit discovers it. Yet on the held-out window it does not beat the global fix on the trade-off that matters: the mean tertile coverage gap is $\overline{\Delta} = 0.073$ for the conditional fit versus 0.071 for the global one, while the conditional fit's median width is larger (Figure 4, Table 2). The reason is sample size: the global multiplier already drags the worst (high-magnitude) tertile from 0.58 to 0.91 on the test split, and the conditional fit's high-magnitude multiplier ($\lambda = 2.02$) is an order statistic of only a few dozen calibration forecasts, so it overshoots to 0.99 rather than landing on 0.90.

We therefore decline to claim the conditional method on this data, and we state the criterion for adopting it: magnitude conditioning should be deployed once the calibration window contains

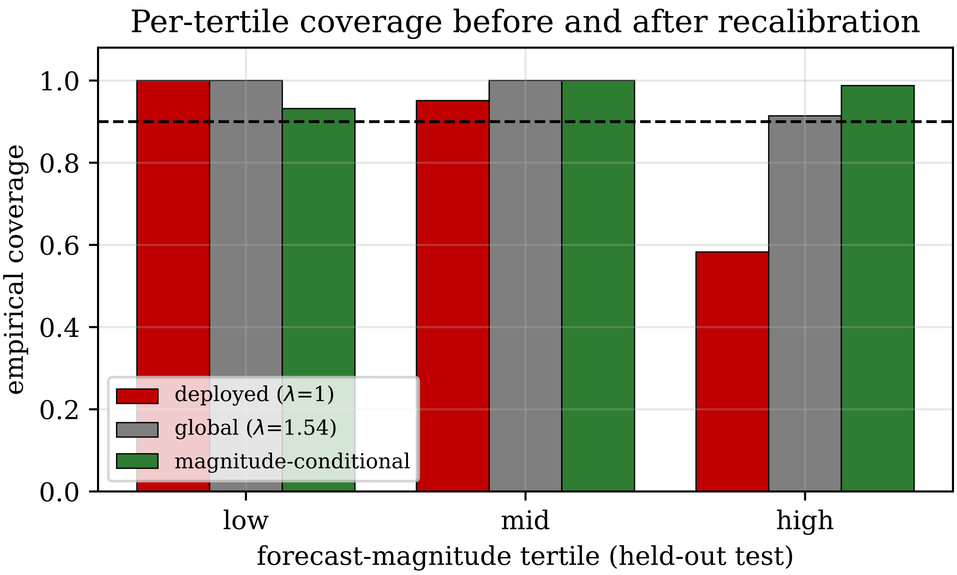

Figure 4: Per-magnitude-tertile coverage on the held-out test window before and after recalibration. The deployed bands under-cover the confident (high-magnitude) tertile at 0.58; the global fix lifts it to 0.91; the magnitude-conditional fit applies the correctly-signed but noisier corrections and overshoots. Dashed line: nominal 0.90.

enough high-magnitude resolved forecasts that the 0.90-quantile of their nonconformity scores is stable — concretely, when the high-magnitude calibration count is large enough that bootstrap resampling of $\lambda^\star_{\text{high}}$ has a coefficient of variation below, say, 0.1. On the present sample it is far above that. The restraint is itself a result: in a thin deployment, a single global recalibration captures most of the recoverable coverage, and conditioning is a structure to grow into, not a knob to turn now.

6.3. Stability over the deployment window

Figure 5 tracks weekly coverage across the deployment. The deployed bands hover well below nominal throughout, with no week reaching 0.90, while the globally recalibrated bands sit around nominal across weeks. The miscoverage is thus a persistent property of the system, not an artifact of one turbulent week — which is what licenses a static multiplicative correction rather than a reactive one. (How quickly such a multiplier should be refreshed as the regime drifts is a separate question, addressed by recalibration-cadence analyses and outside our scope here.)

7. Discussion

Why confidence and coverage invert. The inversion is not a bug peculiar to one system; it is what one should expect whenever a point forecaster operating near the efficiency limit emits intervals whose width is decoupled from its own conviction. Near efficiency, large predicted moves are not more likely to be realized; if anything they revert. An interval that widens with horizon and volatility but not with the forecast's own magnitude will therefore cover the timid forecasts (whose small predicted moves sit comfortably inside a horizon-scaled band) and fail the confident ones (whose large predicted moves push the center away from where the price actually goes, while the band stays put). The remedy is to tie interval width to the quantity that actually predicts miss size — here, forecast magnitude — rather than to proxies like width or horizon that do not. This is a design principle, not a parameter: *an interval should widen in step with the boldness of the point it surrounds.*

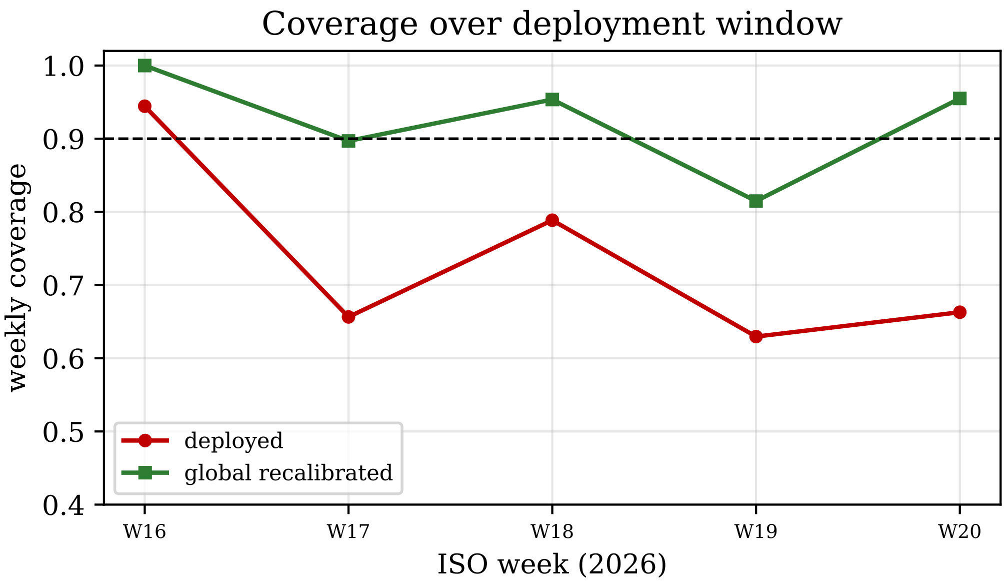

Figure 5: Weekly empirical coverage over the deployment window. Deployed intervals (red) never reach nominal; globally recalibrated intervals (green) track 0.90.

Coverage is necessary, not sufficient. We have been careful to evaluate the correction with a proper score, because coverage alone is a corruptible target: any band can be made to cover by inflating it. The interval-score improvement ($0.218 \to 0.185$) certifies that the conformal widening buys coverage at an honest price in sharpness. Practitioners studying their own systems should report both, and should treat a coverage fix that worsens a proper score as a warning, not a success.

The point is not the product; the interval is. The forecasts studied here carry no directional edge — realized sign matches forecast sign 45.7% of the time, below a coin. It would be easy to read this as a failure of the system. We read it differently: in a near-efficient market the honest deliverable is calibrated uncertainty, and a system that abstains from directional commitment on 94% of cases (its bands straddle the anchor) and quotes a trustworthy band on the rest is behaving correctly — *provided the band is trustworthy.* That proviso is the whole point of this paper, and it is exactly the property that was broken and is now corrected.

8. Threats to Validity and Reproducibility

Sample scope. The analysis covers six instruments over about eight weeks in a single market regime. The numbers characterize this deployment, not equities at large; coverage, the multiplier $\lambda^\star$, and the magnitude structure could differ in another regime. We have therefore framed every quantitative claim as a property of the ledger and have avoided extrapolation. The qualitative result — that miscoverage is indexed by forecast magnitude, not width — is mechanistic and we expect it to recur wherever interval width is decoupled from forecast conviction, but that is a hypothesis for replication, not a proven generalization.

Exchangeability. The split-conformal guarantee assumes exchangeability between calibration and test forecasts. Markets drift, so the assumption is approximate; this is precisely why we report walk-forward coverage on a future window rather than relying on the in-sample guarantee, and why the temporal-stability check matters.

Reproducibility. The analysis is a deterministic function of an immutable ledger. Given the ledger, every figure and table in this paper is regenerated by a single script that recomputes coverage, the nonconformity scores, the conformal multipliers under the fixed 60/40 time split, and the interval scores. No randomness enters except the fixed split point, which is determined by issue-time ordering. The nonconformity score, the conformal quantile rule, and the interval score are stated in closed form in Sections 5. We regard the write-once, point-in-time ledger as the reusable artifact: any team that records forecasts this way can run the same analysis on its own system.

9. Conclusion

We studied the prediction intervals of a deployed equity forecaster against an append-only, leakage-free ledger and found them overconfident (70.6% coverage for a nominal 90% band). The substantive finding is structural: miscoverage is governed not by interval width — coverage is flat across width quartiles — but by the magnitude of the point forecast, collapsing from 95.7% to 47.1% as the model grows more opinionated, because the bands fail to widen for confident forecasts. A one-parameter split-conformal width recalibration estimated on past forecasts restores coverage to 94.4% walk-forward and improves a proper interval score, while a magnitude-conditional refinement recovers the correct corrections but is not yet justified by the sample. The practical message is compact: in a near-efficient regime, *confidence is not coverage*, and an interval keeps its promise only if its width scales with the boldness of the point it surrounds. The evaluation procedure — a write-once ledger, a magnitude-indexed coverage diagnosis, and a conformal correction scored by a proper rule — transfers to any deployed probabilistic forecaster.

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